



Different types

Squares and rectangles are the most obvious quadrilaterals, but there are others too. Here are the 6 main types:

SQUARE

RECTANGLE

RHOMBUS

PARALLELOGRAM

TRAPEZIUM

KITE

SHAPES *with*

What do windows, walls, doors, the pages of this book, and millions of other man-made objects have in common? All of them are rectangles.

Rectangles and other four-sided shapes are everywhere because they're easy to make and fit together neatly. Leave the book for a moment and look around you – how many can you count?

- 1 Draw a quadrilateral with a ruler. Cut it out and tear off all four corners.



- 2 Turn the corners round and see how they fit together.

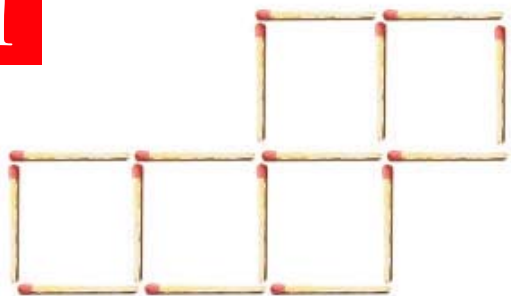


Four angles

The four corners of a quadrilateral will always fit together perfectly, proving that the four corner angles always add to 360° (one whole revolution). If you remember from the previous page, a triangle's angles add up to 180° . So a quadrilateral is like two triangles added together.

PUZZLES

1



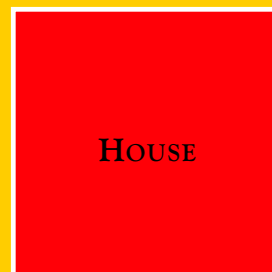
Arrange 16 matches in the pattern above. Can you figure out how to move *only two matches* so that there are four squares instead of five? You can't remove any matches and you can't leave any loose ends.

2

The owner of a square house wants to double the size of his home while still maintaining its square shape. There are four trees near the corners of the house, and the owner can't move them. He doesn't want to build a new storey or a basement, so how can he do it?



TREE



HOUSE